

Attribution-quality metrics: MaRC and IG, baseline vs ABTT

200 POSITIVE test pairs per model (same-folder pairs only; harder/more meaningful slice than the 200-random set); LaTa + PhilTa.

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Method	Suff@25% $\uparrow$		Comp@25% $\uparrow$		Cmpct@0.8 $\downarrow$		ρ_{LOO} $\uparrow$	
	base	abtt	base	abtt	base	abtt	base	abtt
<i>LaTa</i>								
IG	0.521	0.735	0.036	0.613	0.359	0.363	-0.055	0.460
MaRC	0.039	0.740	-0.019	0.564	0.905	0.393	-0.124	0.455
<i>PhilTa</i>								
IG	0.807	0.512	0.176	0.416	0.132	0.501	0.058	0.245
MaRC	-0.350	0.424	-0.002	0.430	0.648	0.618	-0.080	0.295

Within each (method, metric) cell, the better of {baseline, ABTT} is **bolded**. Group means over 200 positive (same-folder) test pairs per model. **Cosine note:** ABTT `full.cos.mean` is now *below* baseline (0.564 vs 0.934 for LaTa; 0.574 vs 0.954 for PhilTa), reversing the direction-of-cleaning bug from the 200-random run (where ABTT was 0.985 > baseline 0.926); per the cosine-inflation investigation, values remain elevated above the D=10 refit demo range (~ 0.30), so this is a partial fix.

Per-cell ABTT-wins count (out of 4 metrics):

- LaTa $\times$ IG: **3/4** (suff + comp + rho; cmpct essentially tied at 0.359 vs 0.363)
- LaTa $\times$ MaRC: **4/4** (clean sweep; cmpct collapses from 0.905 to 0.393)
- PhilTa $\times$ IG: **2/4** (comp + rho; baseline already strong on suff and cmpct)
- PhilTa $\times$ MaRC: **4/4** (clean sweep)

Reading guide. **Suff@25%** ($\uparrow$): similarity retained with only the top 25% attributed tokens. **Comp@25%** ($\uparrow$): similarity drop when those tokens are removed. **Cmpct@0.8** ($\downarrow$): token fraction to recover 80% of full-pair cosine. **ρ_{LOO}** ($\uparrow$): rank correlation between attribution magnitude and true leave-one-out $\Delta \cos$.

Bottom line at 200 *positive* pairs/model.

- The “ $\geq 3/4$ metrics for both MaRC and IG on at least one T5 model” threshold is now **cleared on 3 of 4 cells** (LaTa $\times$ IG 3/4, LaTa $\times$ MaRC 4/4, PhilTa $\times$ MaRC 4/4). On the 200-random set it cleared on 0/4. The positive-pair slice tells a substantially stronger story.
- ρ_{LOO} now improves on **all 4** T5 cells (range +0.19 to +0.58). The “ABTT universally lifts ρ_{LOO} ” claim from the original 20-pair snapshot *is* supported on this slice; the regression seen at PhilTa $\times$ MaRC on the random set ($-0.036 \rightarrow -0.120$) does not appear here ($-0.080 \rightarrow +0.295$).
- The holdout is PhilTa $\times$ IG: ABTT loses Suff@25% ($0.807 \rightarrow 0.512$) and Cmpct@0.8 ($0.132 \rightarrow 0.501$). The PhilTa baseline already retains 81% of the cosine with only the top quartile of tokens, and recovers 80% of full-pair cosine using only 13% of tokens — a hard ceiling for ABTT to clear.
- Caveat unchanged: cross-variant deltas conflate two effects (ABTT changes both the attribution scores *and* the cosine being explained, and the cosine is now meaningfully lower under ABTT — 0.56–0.57 vs 0.93–0.95), so these are descriptive shifts on different fidelity targets rather than apples-to-apples.